

B5-BEYOND-LAYER-BOUND claim-level synthesis: the beyond-layer machinery (DR-2, STEP-5B, SC-SCOPE)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-12 · this version issued 2026-06-13 · v1.4

1. Purpose and scope

Claim-level synthesis for B5-BEYOND-LAYER-BOUND. It adds no mathematics; every statement carries the unit's registered tier. **Scope:** the composition of B5's units into the canonical claim state, and the precise tier relations between the claim (now T7-SCOPE_{admissibility-bounded}) and its units. **Not claimed:** the vacuum selection itself (that is B1/B2's head), arbitrary non-lattice DR-2, or full STEP-5B exhaustiveness beyond the amended-class pin.

2. The claim and its canonical state

Claim (canonical, post-T-031). Pattern-generic Gershgorin reduction of the beyond-layer bound: T7-SCOPE on the admissibility-bounded statement, NOT (the historical T6-conditional form was on $\mathcal{H}_{B5}^{T6} = \{\text{lattice-class H-ADM-COH discharged; non-lattice remainder excluded (T-030); SC-SCOPE endpoint thinness accepted; sunset-limited joint } \times 1.040 > 1 \text{ accepted}\}$; scope CLOSED@LATTICE-DISCHARGED + NONLATTICE-REMAINDER; label B5-BeyondLayer-T6Conditional-260612). The CURRENT canonical state (2026-06-13) is **T7-SCOPE**_{admissibility-bounded} given A1-KERNEL-CONV (label B5-BeyondLayer-T7Scope-260613): for every admissible competitor the beyond-layer correction is below the layer margin, with NO substantive conditional hypothesis – H-ENDPOINT-THINNESS-ACCEPTED removed (sunset hardening) and H-NONLATTICE-REMAINDER-EXCLUDED reclassified to definitional scope (route-3; Lemma 2 caps $T' \leq 10$ pattern-generically). **Mandatory qualifier:** B5 does NOT claim unrestricted arbitrary- Q DR-2; T-030 stays OPEN as a frontier strengthening. (Historical: T5 PINNED-CLOSURE 2026-06-06; T6-conditional 2026-06-12; T7-SCOPE 2026-06-13.)

3. Sub-proof unit map (per-unit tier and role)

Unit	Tier	What it contributes	Publication state
DR-2/	T7 (lattice, std. NT)	Lemma 1 backing: $E_+(Q) \leq_\epsilon R^\epsilon N^2$ decoupling-free on lattice shells; Lemma NT pinned in-bundle	PUBLISHED – DR2-Lattice-T7-NTstandard-260612 (8/8, 38/38)
STEP-5B/	T6	the rectangle prefactor $K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n} + \text{official threshold } n_{\text{adm}} \lesssim 1.59 \times 10^5$	PUBLISHED – STEP-5B-Rectangle-T6-260612 (192/192)
SC-SCOPE/	T6 (sunset-hardened)	the (S) selection floor at the endpoint: joint = $\times 2.023$ (cons.) / $\times 1.886$ (worst dressing) > 1 on $\mathcal{W}_{\text{SC}}$ – per-transfer sunset hardening retired the thinness	PUBLISHED – SC-SCOPE-SunsetHardened-T6-260612 (5/5, 39/39)
H-LAYER-AUX/	T5	RES-4 layer-ratio positivity $\rho(I) \geq \times 2.58 > 1$, monotone (201 nodes) – input to STEP-5B's K-budget	AUXILIARY (operator-accepted v1.1)
T5-DOSSIER/	T5	the assignment dossier pinning the claim tier (closure pillars, 192/192)	internal consolidation

4. Composition: the claim tier vs its units (T7-SCOPE, 2026-06-13)

The claim’s statement is the class-scoped closure; its strongest units exceed it where applicable (DR-2 T7-NTstandard; STEP-5B rectangle T6; SC-SCOPE T6 sunset-hardened) and are cited at their own tiers. The upgrade path foreseen here (the T-031 exhaustiveness decision layer + the R-U6-1 backlog lemma) was EXECUTED on 2026-06-12: R-U6-1/2 discharged, T-031 verdicts D1-A/D2-A/D3-A enacted, and the claim re-tiered to T7-SCOPE on the admissibility-bounded statement, the substantive hypothesis set EMPTIED (the budget comparison was already machine-closed at $\times 55.6 / \times 8.8 / \times 2.1 - 2.6$). **The former T7 path has been EXECUTED (2026-06-13):** both acceptance hypotheses are disposed – H-ENDPOINT-THINNESS-ACCEPTED REMOVED (per-transfer sunset hardening, SC-SCOPE-SunsetHardened-T6; endpoint joint $\times 1.886 - \times 2.023 > 1$) and H-NONLATTICE-REMAINDER-EXCLUDED reclassified to DEFINITIONAL SCOPE (route-3: Lemma 2 caps $T' \leq 10$ pattern-generically for all admissible competitors, (T', n) -only Attack-4 audit 6/6). B5 is now T7-SCOPE_{admissibility-bounded}; T-030 (unrestricted arbitrary- Q) stays OPEN as a frontier strengthening, non-load-bearing for B5’s canonical statement.

Consumption map (who uses what): B1/B2’s head uses Lemma 1 (DR-2 mechanism) + Lemma 2 (in-head packing) for (O), SC-SCOPE for (S), STEP-5B for the K-budget; H-LAYER-AUX feeds STEP-5B. Per-edge accounting: parent capstone Main-Line-Synthesis-T013-260612 Sec. 3–4.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*Admissibility hides a hypothesis – this is really T6-conditional.*” ADDRESSED: the $\theta_{\min}$ coherence-resolution admissibility is DEFINITIONAL SCOPE (the physical competitor class, fixed by the microphysics), not a substantive unproved hypothesis – exactly as ‘ $\mathcal{C}_{\text{full}}$ ’ is the definitional scope of the B1/B2 head’s own T7-SCOPE. A scoped-T7 with empty substantive conditions is T7-SCOPE, not T6-conditional (which requires a NAMED unproved hypothesis; there is now none).
- (β) “*A1-KERNEL-CONV remains a dependency.*” VALID and explicit: A1 is the named DEFINITIONAL input (the kernel convention), carried in the scope, identical to the B1/B2 head – B5 is T7-SCOPE GIVEN A1, a definition, not a conditional hypothesis.
- (γ) “*T-030 is open – does it conflict with B5 T7-SCOPE?*” DISMISSED: T-030 (unrestricted arbitrary- Q DR-2) is a FRONTIER STRENGTHENING that removes the admissibility cap; it is non-load-bearing for B5’s admissibility-bounded statement (route-3, (T', n) -only audit). B5 T7-SCOPE and T-030 OPEN coexist exactly as B1 T7-SCOPE and the open global/all-intensity questions coexist.

Quantitative sanity check (mandatory). Cross-unit consistency: the STEP-5B rectangle bound at the class size ($K(n_{\text{pack}}) \leq 113$) vs the measured $K = 107$ (6%, safe side); $T' \leq 10$ (Lemma 2) gives $R_{\text{lead}} = 0.510 < 1$; the sunset-hardened (S) joint is $\times 2.023$ (conservative) / $\times 1.886$ (worst mixed dressing) > 1 ; all current bundle digests verified at publication.

6. Result footer

Result ID: B5-BEYOND-LAYER-BOUND-Synthesis-T013-260613
(PUBLISHED claim-level SYNTHESIS, T-013)

Precise statement: B5’s canonical state is
T7-SCOPE_{admissibility-bounded} given
A1-KERNEL-CONV (label
B5-BeyondLayer-T7Scope-260613); substantive
hypotheses NONE. Its sub-proof units hold
PUBLISHED bundles at their own tiers (DR-2
T7-lattice, STEP-5B T6, SC-SCOPE T6

	sunset-hardened) and are cited at those tiers. Claim tier != max unit tier.
Scope:	admissibility-bounded; NOT unrestricted arbitrary-Q. Synthesis layer; no new math.
Evidence grade:	SYNTHESIS over the PUBLISHED bundles + operator-accepted auxiliaries.
Reproduction command:	run the published bundles' entry manifests.
Falsifier:	any falsifier of the cited footers; or a unit-tier mismatch against INDEX.md.
Tier before / after:	B5 T6 PROVED-CONDITIONAL -> T7-SCOPE_{admissibility-bounded} (route-3 promotion 2026-06-13); this note SYNCs the synthesis to that state (no tier action here).
No-overclaim statement:	B5 T7-SCOPE is admissibility-bounded, given A1-KERNEL-CONV; T-030 remains OPEN as a frontier strengthening, non-load-bearing for B5 -- NOT an unrestricted/global/arbitrary-Q claim.
Next required action:	none for this synthesis (freeze-clean); optional claim-card/status sync only.